ADULT ECHOCARDIOGRAPHY SONOGRAPHY SYLLABUS

COURSE CODE: DMS 500

ADULT ECHOCARDIOGRAPHY SONOGRAPHY

Instructor:
Telephone:
E-Mail:
Office Hours:
Mon, Wed, Fri: 12pm to 1pm & 5pm to 6pm
Tues: 9am – 5pm
Thurs: 11am – 5pm

Start Date:

Completion Date:

Days and Times:
Monday, Wednesday and Friday

Course Hours:
Morning Session:
8:00 am -10:00am &
10:10am – 12:00 pm
Afternoon Sessions:
1:00pm – 3:00pm &
3:10 pm – 5:00 pm

Class Location:
Classroom #:
Lab#:

Prerequisites:
BIO 100 Introduction to A&P
PAT 100 Patient Care
LAW 100 Medical Law & Ethics

Course Contact Hours: 292 hours
Lecture: 116 hours
Laboratory: 176 hours
Course Delivery: Residential
Pre-requisite: None

Americans with Disabilities Act Guidelines:

Requests for accommodations should be submitted to the campus president. Faculty are not authorized to grant accommodations.

COURSE DESCRIPTION

This introductory level course is directed towards developing the principals, techniques, skills and fundamental knowledge of each student necessary to function as a qualified entry level diagnostic medical cardiac sonographer.
It introduces the foundations of diagnostic medical sonography as it pertains to the echocardiography images, including medical terminology related to the heart anatomy, physiology and pathology. Students will learn hemodynamic principals and scanning plane orientations (longitudinal, transverse and coronal) related to echocardiography.

Emphasis will be placed on the clinical applications of echocardiography, including basic patient care principals, professionalism, clinical signs and symptoms. Scanning protocols and techniques will be presented in a demonstrated lesson in conjunction with laboratory scanning sessions.
The academic laboratory and clinical courses are integrated into the program for students to learn and practice different echocardiographic scanning protocols.
Interpretation of normal and pathologic echocardiography patterns will be learned including the normal variants, laboratory tests and values. At the end of this curriculum, students will possess knowledge and skills necessary for employment in the healthcare system as an entry level echocardiographer sonographer.

Instructional Materials

1. Textbook
Title: Textbook of Diagnostic Sonography, 8th Edition (2 Volume Set)
Author(s): Sandra L. Hagen-Ansert, MS, RDMS, RDCS, FASE, FSDMS
Publisher: Mosby; 8th edition (May 3, 2017)
ISBN: 978-0323353755

2. Workbook
Title: Workbook for Diagnostic Sonography, 8th Edition
Author(s): Sandra L. Hagen-Ansert, MS, RDMS, RDCS, FASE, FSDMS
Publisher: Mosby; 8th edition (May 3, 2017)
ISBN: 978-0323441834

3. Reference Book
Title: Pocket Protocols for Sonography Scanning, 4th Edition
Author(s): Betty Bates-Tempkin, BA, RT®, RDMS
Publisher: Saunders; 4th edition (December 27, 2014)
ISBN: 978-1455773220

4. Textbook
Title: ASE Comprehensive Echocardiography, 2nd Ed.
Author(s): Lang Goldstein
Publisher: Elsevier
ISBN: 978-0323260114

5. Textbook
Title: Feigenbaum's Echocardiography, 8 ed
Author(s): William F. Armstrong, Thomas Ryan, M.D.
Publisher: LWW
ISBN: 978-1451194272

Equipment/Technology

- LCD Projector
- Laptop Computer
- Mindray Ultrasound Machine

Material

- Electronic and Printed Worksheets
- Textbooks
- Visual Aids: DVD Videos, Case Study Reviews, Demonstrations and Hands on Clinical Laboratory Practice.

During clinical laboratory sessions various techniques such as role playing and scanning exercises will be used.

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Grade Point Value

Grade	Percentile Range	Points
A+	97-100	4.00
A	93-96.99	3.85
A-	90. -92.99	3.67
B+	87-89.99	3.33
B	83-86.99	3.00
B-	80-82.99	2.67
C+	77-79.99	2.33
C	73-76.99	2.00
C-	70-72.99	1.67
D+	67-69.99	1.33
D	63-66.99	1.00
D-	60-62.99	.67
F	Below 60-00	0

Methods of Evaluation

Exams / Quizzes/ Midterm/ Final	40%
Projects/ Attendance Participation/ Homework Assignments	20%
Scan-out	40%
	100%

Teaching Strategies

A variety of teaching strategies may be utilized in this course, including but not limited to: lecture, discussion, written classroom exercises, visual aids, videos, case study reviews, group and/or individual projects, quizzes, exams, reading assignments, writing assignments, interactive exercises demonstrations and hands, internet research and hands on clinical laboratory practice.

Examination Standards

Students are expected to be honest in all academic work. Cheating of any kind is not tolerated. Violations of this principle will result in a grade of zero in the course and additional disciplinary action including expulsion as determined by CURE.

COURSE OBJECTIVES

Upon the successful completion of this courses the student should be able to:

- Describe and analyze the anatomy of cardiac structures and associated vasculature.
- Explain the transducer choice and patient position for a cardiac examination.
- Describe the imaging planes used in echocardiography.
- Define parasternal apical, subcostal, and suprasternal.
- Describe a normal cardiac examination using two-dimensional, color flow, Doppler, and M-mode imaging modes.
- Recognize normal and pathologic echocardiographic appearances.
- Find the echocardiographic distinction between pericardial effusion and pleural effusion.
- Describe the echocardiographic findings in cardiac tamponade.
- Describe the echocardiographic findings in dilated, restrictive, infiltrative, and hypertrophic cardiomyopathy.
- Recognize the echocardiographic findings of tumors, thrombus, vegetations, and normal variant.
- Describe the characteristics of the spectral doppler waveforms obtained during echocardiography scanning.
- Discuss common errors associated with performing and interpreting echocardiographic imaging.
- Record and save video clips and images of the normal or pathologic echocardiographic appearance pattern.
- Pass the didactic course with a minimum passing grade of 73.
- Pass the scanning laboratory echocardiography scanning protocols.

Attendance Policy

Routine, regular class attendance is a requirement for this course.
Promptness and dependability are the soft skills that garner special attention

Attendance Rubric:

Up to 25% will be deducted for each hour missed or for any part thereof (5% for each 10 minutes missed). Students are required to meet their professor to discuss matters of tardiness or absence.

Students are required to notify their instructor via E mail or voicemail to communicate an absence or lateness from class

A student absent from all of his or her classes for a period of 14 consecutive calendar days during which classes or examinations are in session will be regarded as having withdrawn from the program and will receive a grade of W for each course in which he or she is enrolled.

HOMEWORK

An essential part of every program of study, homework enables students to master course learning objectives and leads toward the achievement of the overall program objectives. Homework refers to self-directed learning activities such as reading assignments, research activities and projects, quiz/exam preparation, knowledge checks and self-assessments, etc.

Late Homework/Make-Up Policy

Late homework will be accepted under the following guidelines:

1. It must be turned in within two business days (Monday through Thursday) after the due date.
2. However, homework turned in after the due date will be penalized 10% for each day it is late up to two calendar days after the due date.
3. Homework submitted after three business days after the due date will earn a zero.

Exams and quizzes - Make up of a quiz or an exam will be allowed under the following extenuating circumstances only:

- Documented illness
- Military duty
- Death of a family member
- Court appearance or jury duty

Documentation of the above circumstances must be provided to the Program Director. ***The ability to make-up a quiz or exam must be approved by the Program Director.***

All assignments should be submitted directly to the instructor.

Classroom Policy

Our goal is to build a respectful learning and work environment that allows for positive communication and teamwork. To support your development, all students must abide by academic policies related to attendance, behavior and professional conduct as published in the current institutional catalog. Special attention is granted to the following items:

- There is to be no eating or drinking in any lab/computer class.
- There is to be no eating in ANY class.
- Cellular phones may only be used during class if/when advised by the instructor and required for a learning activity.
- All students must be in full uniform which includes white or black shoes at **ALL TIMES.**
- **There are no rescheduled exams *unless specified by the professor* with advance notification.**

- The determination to accept late homework assignments is made only by the instructor and occurs only in the event of extraordinary circumstances. If the professor does allow for an extension on assignments, there will be an automatic daily reduction by 10 points, each day beyond the original due date.
- All classes are expected to start and end at a predetermined start and end time.

Accommodations for Students with Disabilities

If you have a special learning need, please contact the Program Director. The Program Director and the Campus President will work with the student to identify reasonable accommodations or adjustments necessary to enable him or her to fully participate in the educational processes.

Professional Responsibility

Students are expected to aspire to the same high level of ethics and professionalism that is demanded of registered Ultrasound students. This requires students to conduct themselves as professionals and to fully engage in the learning process by preparing for and participating in all classroom, laboratory and clinical activities.
Attendance is taken each hour of class by the instructor and turned over to the Registrar before the end of the class day. Early departures, tardiest, and class cuts will be recorded in quarter-hour increments and a proportional grade reduction will also be applied. A period of less than 15 minutes will be counted as a quarter-hour of absence. Attendance records are maintained by the Registrar as part of the student's permanent academic record.

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COURSE OUTLINE

Below is the course schedule which identifies lecture topics, assignments and other course activities and the homework required for completion each week.

The course syllabus is a general plan for the course; deviations may be necessary and will be announced.

Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
Week One Objectives: • Introduction to course, class policies, and Syllabus • Describe the landmarks of the thoracic cavity. • Define the relational landmarks of the heart. • Discuss the function of the pericardial sac. • Differentiate the three layers of the heart wall. • Describe the anatomic landmarks of the cardiac chambers, valves and interventricular septum.		
DAY 1. Thorax and thoracic Cavity anatomy Anatomy of the Heart	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 30: Anatomic and Physiologic Relationships within the Thoracic Cavity pages; 809- 816 Chapter 30 homework: Workbook Exercise 1 and Exercise 2 Page 239
DAY 2: Great Vessels Coronary Arteries Cardiac Cycle	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 30: Anatomic and Physiologic Relationships within the Thoracic Cavity page: 812-818 Chapter 30 Homework: Workbook exercise 3 page 240-243
DAY 3: Electrical Conduction system Mechanical Conduction System Electrocardiography	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 30: Anatomic and Physiologic Relationships within the Thoracic Cavity page: 818-820 Chapter 30 Homework: Workbook exercise 4, exercise 5, exercise 6 and exercise 7 pages 244-246

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Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
Week Two Objectives: • Understand the fundamental concepts of hemodynamics as applied to ultrasound. • State basic knowledge of the cardiac cycle, normal intracardiac pressures and volumes, cardiac output, stroke volume, and its mechanisms • Describe the principles of blood flow velocity profiles, how they are displayed, and how they are evaluated by Doppler ultrasound. • Demonstrate knowledge of how to apply Doppler ultrasound to calculate pressures, velocities, and output.		
DAY 4. Principles of Blood Flow Normal Intracardiac Pressures and Volumes Cardiac Output, Stroke Volume, and Its Mechanisms Blood Flow Velocity Profiles QUIZ	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 31: Understanding Hemodynamics pages; 824-828 Chapter 31 Homework: Workbook Exercise 1 page 247
DAY 5: Doppler Basics Doppler Effect and Frequency Shift Pulsed Wave Doppler Color Doppler Continuous Wave Doppler	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 31: Understanding Hemodynamics pages; 828-831 Chapter 31 Homework: Workbook Exercise2 page 248-249
DAY 6: Quantification of Intracardiac Pressures by Ultrasound Pressure-Velocity Relationship: Bernoulli Equation Right Arterial Pressure Cardiac Output and Stroke Volume Measured by Ultrasound Valve area: Continuity Principle and Equation	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 31: Understanding Hemodynamics pages; 828-831 Chapter 31 Homework: Workbook. Exercise 3 page 249-250

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Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
Week Three Objectives: • Explain the transducer selection and patient position for a cardiac examination. • Describe the imaging planes used in echocardiography. • Define suprasternal, subcostal, apical and parasternal. • Describe a normal cardiac examination using two-dimensional, color flow, Doppler, and M-mode imaging modes. • List and discuss the applications of color flow Doppler in the echocardiographic examination. • Analyze comparative anatomy of cardiac structures.		
DAY 7. Two-Dimensional Echocardiography Doppler Application and Technique QUIZ	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 32: Introduction to Echocardiographic Techniques, Terminology, and Tips, pages; 832-838 Chapter 32 Homework: Workbook Exercise 1 page 251
DAY 8: Echocardiographic Examination M-Mode Imaging of the cardiac Structures	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 32: Understanding Hemodynamics pages; 838-861 Chapter 32 Homework: Workbook Exercise 2 page 251
DAY 9: Echocardiographic Examination M-Mode Imaging of the cardiac Structures	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 32: Understanding Hemodynamics pages; 838-861 Chapter 32 Homework: Workbook Exercise 3 pages 252-257

Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
Week Four Objectives: • Define the characteristics and causes of mitral regurgitation. • Know how to calculate the PISA method. • Describe the common causes of mitral stenosis.		
DAY 10. Left-Sided Valvular Heart Disease QUIZ	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 33 Introduction to Echocardiographic: Left-sided Valvular Heart Disease pages; 863-865 Chapter 33 Homework: Workbook Exercise 1 and Exercise 2 pages 259-261
DAY 11: Mitral Valve Disease Mitral Regurgitation	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 33 Introduction to Echocardiographic: Left-sided Valvular Heart Disease pages; 863-871 Chapter 33 Homework: Workbook Exercise 3 page 261
DAY 12: Mitral Stenosis	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 33 Introduction to Echocardiographic: Left-sided Valvular Heart Disease pages; 863-871 Chapter 33 Homework: Workbook Exercise 4 pages 261-266

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Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
Week Five Objectives: • Explain the causes of aortic regurgitation. • List the echocardiographic features of acute versus chronic aortic regurgitation. • State the three types of aortic stenosis and their echocardiographic features. • Explain how to calculate the aortic valve area. • Define aortic aneurysm. • State the difference between a true aneurysm and a pseudoaneurysm. • Describe the echocardiographic finding in aortic dissection.		
DAY 13: Aortic Valve Disease Aortic Insufficiency QUIZ	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 33 Introduction to Clinical Echocardiographic: Left- sided Valvular Heart Disease pages: 871-875
DAY 14: Aortic Insufficiency Aortic Stenosis	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 33 Introduction to Clinical Echocardiographic: Left- sided Valvular Heart Disease Pages: 875-880
DAY 15: Aortic Stenosis Technical Considerations and Pitfalls in Doppler Measurements	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 33 Introduction to Clinical Echocardiographic: Left- sided Valvular Heart Disease Pages: 875-886

Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
Week Six Objectives: • Identify the echocardiographic distinction between pericardial effusion and pleural effusion.		
DAY 16. MID TERM Pericardial Disease	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 34: Introduction to clinical Echocardiography: Pericardial Disease, Cardiomyopathies, and Tumors pages; 887-889 Chapter 34 Homework: Workbook Pages 267-273
DAY 17: Pericardial Disease	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 34: Introduction to clinical Echocardiography: Pericardial Disease, Cardiomyopathies, and Tumors pages; 887-889 Chapter 34 Homework: Workbook Pages 267-273
DAY 18: Pericardial Disease	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 34: Introduction to clinical Echocardiography: Pericardial Disease, Cardiomyopathies, and Tumors pages; 887-890 Chapter 34 Homework: Workbook Pages 267-273

Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
Week Seven Objectives: Describe the echocardiographic findings in dilated, restrictive, infiltrative, and hypertrophic cardiomyopathy.		
DAY 19. Cardiomyopathies QUIZ	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 34: Introduction to clinical Echocardiography: Pericardial Disease, Cardiomyopathies, and Tumors pages; 890- 903 Chapter 34 Homework: Workbook Pages 267-273
DAY 20: Cardiomyopathies	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 34: Introduction to clinical Echocardiography: Pericardial Disease, Cardiomyopathies, and Tumors pages; 890- 903 Chapter 34 Homework: Workbook Pages 267-273
DAY 21: Cardiomyopathies	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 34: Introduction to clinical Echocardiography: Pericardial Disease, Cardiomyopathies, and Tumors pages; 890- 903 Chapter 34 Homework: Workbook Pages 267-273

| Lecture Topic | In-Class Activity/Lab Assignment | Homework/Out of Class |

Week Eight Objectives:

- Describe the echocardiographic findings in dilated, restrictive, infiltrative, and hypertrophic cardiomyopathy.
- Recognize the echocardiographic findings of tumors, thrombus, vegetations, and normal variant.

DAY 22. Cardiomyopathies QUIZ	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 34: Introduction to clinical Echocardiography: Pericardial Disease, Cardiomyopathies, and Tumors pages; 890- 903 Chapter 34 Homework: Workbook Pages 267-273
DAY 23: Cardiac Masses	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 34: Introduction to clinical Echocardiography: Pericardial Disease, Cardiomyopathies, and Tumors pages; 903-909 Chapter 34 Homework: Workbook Pages 267-273
DAY 24: Cardiac Masses	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 34: Introduction to clinical Echocardiography: Pericardial Disease, Cardiomyopathies, and Tumors pages; 903-909 Chapter 34 Homework: Workbook Pages 267-273

Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class

Week Nine Objectives:

Understand the embryologic development of the fetal heart.

- Describe fetal circulation.
- List the material risk factors that would require fetal echocardiography.
- Explain how to sonography evaluate the fetal heart with two- dimensional, M-Mode, pulsed Doppler, and color flow Doppler imaging.
- Locate fetal ultrasound landmarks/Describe normal anatomy seen in the views discussed in this chapter.

DAY 25. Cardiac Masses QUIZ	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 34: Introduction to clinical Echocardiography: Pericardial Disease, Cardiomyopathies, and Tumors pages; 903-909 Chapter 34 Homework: Workbook Pages 267-273
DAY 26: Review Fetal Echocardiography	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 35: Fetal Echocardiography: Beyond the Four Chambers pages; 910- 926 Chapter 35 Homework: Workbook Pages Exercise 1, Exercise 2 & Exercise 3 page 275-277
DAY 27: Review Fetal Echocardiography	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 35: Fetal Echocardiography: Beyond the Four Chambers pages; 910- 926 Chapter 35 Homework: Workbook, Exercise 4, Exercise 5 & Exercise 6 pages 278-282

Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class

Week Ten Objectives:

- List the three factors that contribute to congenital heart disease.
- Describe why the four-chamber view cannot rule out all forms of congenital heart diseases.
- Discuss the pathologic conditions covered in this chapter.
- Discuss the echocardiographic findings for septal defects, ventricular inflow and outflow tract disturbances, great vessel abnormalities, cardiac tumors, complex cardiac abnormalities, and dysrhythmias.

DAY 28. Congenital Heart Disease QUIZ	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 36: Fetal Echocardiography: Congenital Heart Disease pages; 927-964 Chapter 36 Homework: Workbook Exercise 1 page 283
DAY 29: Congenital Heart Disease	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 36: Fetal Echocardiography: Congenital Heart Disease pages; 927-964 Chapter 36 Homework: Workbook Exercise 2 page 284
DAY 30: Congenital Heart Disease	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 36: Fetal Echocardiography: Congenital Heart Disease pages; 927-964 Chapter 36 Homework: Workbook Exercise 3 page 285

Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
Week Eleven Objectives: • List the three factors that contribute to congenital heart disease. • Describe why the four-chamber view cannot rule out all forms of congenital heart diseases. • Discuss the pathologic conditions covered in this chapter. • Discuss the echocardiographic findings for septal defects, ventricular inflow and outflow tract disturbances, great vessel abnormalities, cardiac tumors, complex cardiac abnormalities, and dysrhythmias.		
DAY 31. Congenital Heart Disease QUIZ	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 36: Fetal Echocardiography: Congenital Heart Disease pages; 927-964 Chapter 36 Homework: Workbook Exercise 4 page 285-287
DAY 32: Congenital Heart Disease	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 36: Fetal Echocardiography: Congenital Heart Disease pages; 927-964 Chapter 36 Homework: Workbook Exercise 5 page 287
DAY 33: Congenital Heart Disease	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 36: Fetal Echocardiography: Congenital Heart Disease pages; 927-964 Chapter 36 Homework: Workbook Exercise 6 pages 288-292

| Lecture Topic | In-Class Activity/Lab Assignment | Homework/Out of Class |

Week Twelve Objectives: • List the three factors that contribute to congenital heart disease. • Describe why the four-chamber view cannot rule out all forms of congenital heart diseases. • Discuss the pathologic conditions covered in this chapter. • Discuss the echocardiographic findings for septal defects, ventricular inflow and outflow tract disturbances, great vessel abnormalities, cardiac tumors, complex cardiac abnormalities, and dysrhythmias. • Review.		
DAY 34. Congenital Heart Disease	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 36: Fetal Echocardiography: Congenital Heart Disease pages; 927-964 Chapter 36 Homework: Workbook Exercise 6 pages 288-292
DAY 35: Congenital Heart Disease	PowerPoint Presentation/ Lecture/LAB	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 36: Fetal Echocardiography: Congenital Heart Disease pages; 927-964 Chapter 36 Homework: Workbook Exercise 6 pages 288-292
DAY 36: Review	PowerPoint Presentation/ Lecture/LAB	**Review**

Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
Week Thirteen Objectives: • Final Exam and scan outs.		
DAY 37. Scan-Out/Final	PowerPoint Presentation/ Lecture/LAB	Final Exam